

Regional Stress Response to Macroeconomic Shocks and Earthquakes in Türkiye

A data-driven analysis using Google Trends, earthquake records, FX volatility, and unsupervised machine learning (2018–2025)

Project Repository: <https://github.com/aycaaydogdu/regional-stress-response-turkey>

Ayça Sudem Aydoğdu

Executive Summary

This project studies how Türkiye's seven official geographical regions respond to nationwide stressors—primarily major earthquakes and foreign-exchange (FX) shocks—using Google Trends searches as a proxy for collective stress. The pipeline collects weekly Google Trends interest for five stress-related keywords across all 81 provinces (2018–2025), aggregates them into a regional stress index, detects event dates from earthquake and USD/TRY data, and evaluates changes in stress around each event.

An event-window design compares regional stress levels in the 30 days before and after each event, producing a delta-stress metric (post minus pre). A one-way ANOVA test suggests that observed regional differences in delta-stress are not statistically significant at the 0.05 level ($p = 0.0886$), indicating a largely synchronized national stress response rather than strongly region-specific patterns.

For Phase 3, the project applies K-Means clustering on engineered regional features (mean stress, maximum stress, and stress variability) to explore whether regions form latent “stress-response profiles.” This unsupervised step complements hypothesis testing by providing an interpretable grouping of regions based on overall behavior, even when strict statistical significance is not found.

Document structure: The report covers motivation and research questions, data sources, methodology and pipeline, exploratory analysis and visuals, statistical testing, machine learning clustering, limitations, and future extensions.

1. Motivation and Research Questions

Nationwide shocks such as currency volatility and major earthquakes can trigger rapid changes in public sentiment. However, regional socioeconomic conditions, information diffusion, and local exposure may lead to heterogeneous responses. This project asks whether Türkiye's regions exhibit different stress reaction intensity and timing when faced with the same national stressors.

Research Questions

- Do Türkiye's seven regions show different average stress responses to nationwide events (earthquakes and FX shocks)?
- Are stress responses synchronized across regions, or do some regions consistently react more strongly or more quickly?
- Can unsupervised learning discover latent regional profiles based on stress intensity and variability?

2. Data Sources

The project integrates multiple data streams to relate public stress to event timing. Google Trends provides a near real-time proxy for stress-related public interest, while event calendars are derived from earthquake catalogs and currency market volatility.

2.1 Google Trends (Stress Proxy)

- Coverage: 81 Turkish provinces aggregated into 7 geographical regions.
- Frequency: weekly (2018–2025).
- Keywords: “anksiyete” (anxiety), “uykusuzluk” (insomnia), “stres” (stress), “panik atak” (panic attack), “mide yanması” (heartburn).
- Collection tool: PyTrends (Google Trends API wrapper).
- Output file: google_trends_province_timeseries.csv.

2.2 Event Data

- Earthquakes: events filtered by magnitude ≥ 4.5 and depth between 0–40 km (catalog stored as earthquake.csv).
- FX shocks: dates when daily USD/TRY log returns exceed a 2% threshold (exchange-rate file stored as USD_TRY Historical Data.csv).
- Output file: event_dates.csv with a unified calendar and event type labels.

3. Methodology and Analysis Pipeline

The project is implemented as a reproducible Python pipeline. Each stage produces intermediate CSV outputs to support transparency and re-running the analysis.

3.1 Pipeline Overview

- collect_trends_provinces.py → downloads weekly Google Trends by province and keyword.
- build_region_stress_index.py → constructs region-level stress index via z-score normalization and aggregation.
- build_event_dates.py → detects earthquake and FX shock event dates and merges into a unified event calendar.
- analyze_region_stress.py → computes event-window statistics and delta-stress; runs ANOVA for regional differences.
- plot_region_stress_analysis.py → generates summary plots for trends, delta-stress distributions, and event comparisons.

3.2 Stress Index Construction

Because Google Trends values are normalized and not directly comparable across separate queries, the project standardizes each province–keyword series using z-scores. These standardized values are then averaged across provinces within each geographical region, producing a weekly regional stress index. Higher values represent higher-than-usual search interest in stress-related terms relative to each province’s own baseline.

4. Exploratory Data Analysis

Weekly regional stress indices provide a first look at synchronization versus divergence across the seven regions. Visual inspection highlights periods where stress increases broadly across regions, suggesting national-level drivers.

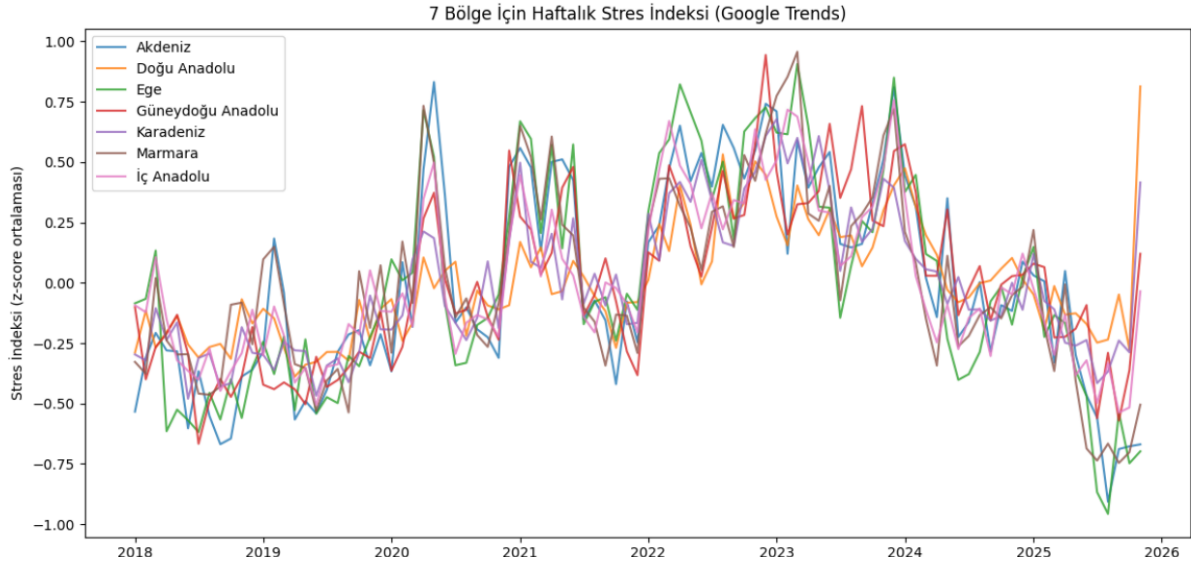


Figure 1. Weekly regional stress index (z-score averaged across stress-related Google Trends keywords), 2018–2025.

Across 2018–2025, regions frequently move together, with shared spikes around major national periods. While the lines occasionally diverge, the overall pattern suggests substantial co-movement, consistent with stressors that affect the whole country.

5. Event-Window Analysis: Measuring Stress Response

To quantify event-driven stress changes, the analysis uses a symmetric event window. For each event date, regional stress is averaged over the 30 days before (pre) and 30 days after (post). The main response metric is:

$\text{Delta Stress} = \text{Post-Event Mean Stress} - \text{Pre-Event Mean Stress}.$

Positive values indicate higher stress after the event; negative values indicate a decrease.

5.1 Average Response by Region

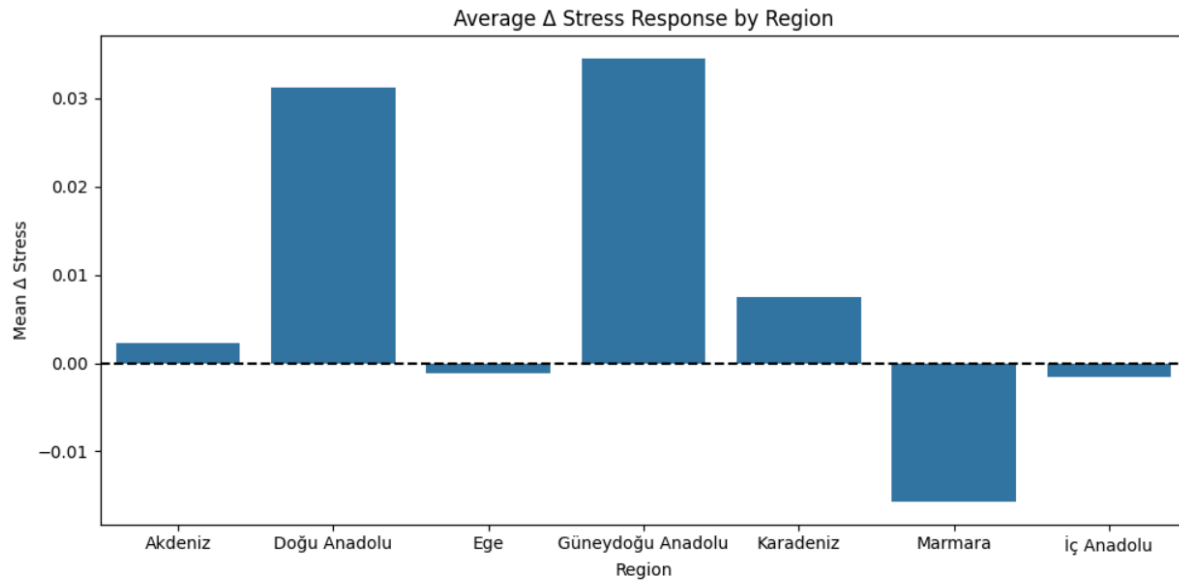


Figure 2. Mean delta-stress (post – pre) by region across all events (earthquakes and FX shocks).

Average delta-stress differs modestly across regions. Some regions show positive average changes while others are near zero or slightly negative. However, averages alone can be misleading when the underlying distribution has wide variance or overlapping ranges.

5.2 Distribution of Responses

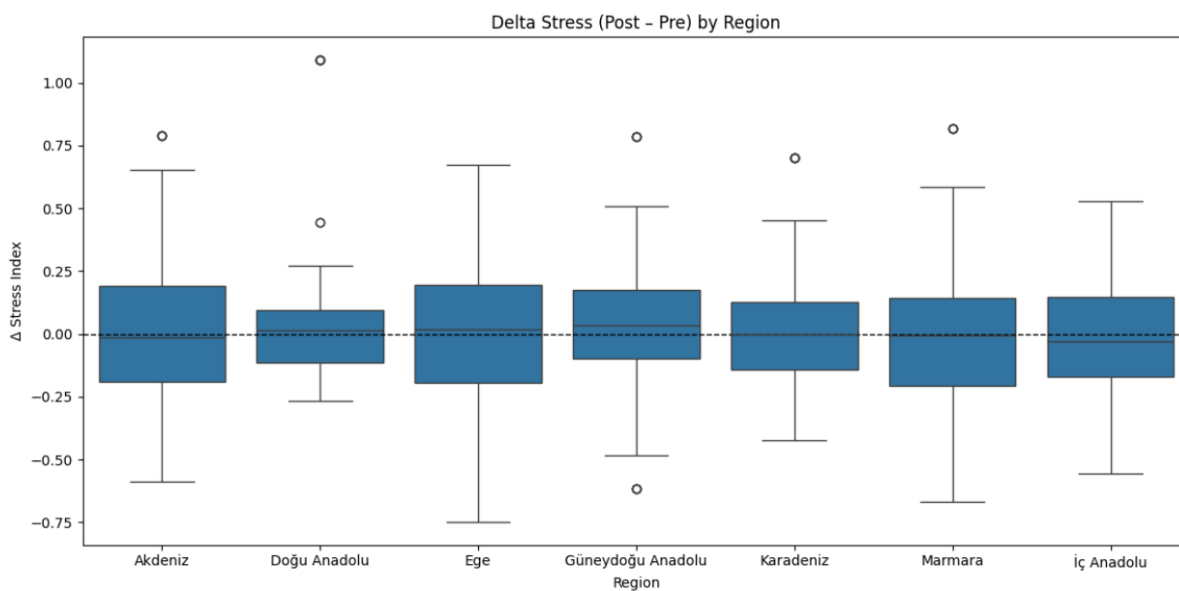


Figure 3. Distribution of delta-stress by region. The dotted line marks zero (no change).

The boxplots show substantial overlap in delta-stress across regions. This overlap suggests that, while individual events can produce large stress changes, these changes do not consistently separate regions into distinct response regimes.

5.3 Earthquakes vs FX Shocks

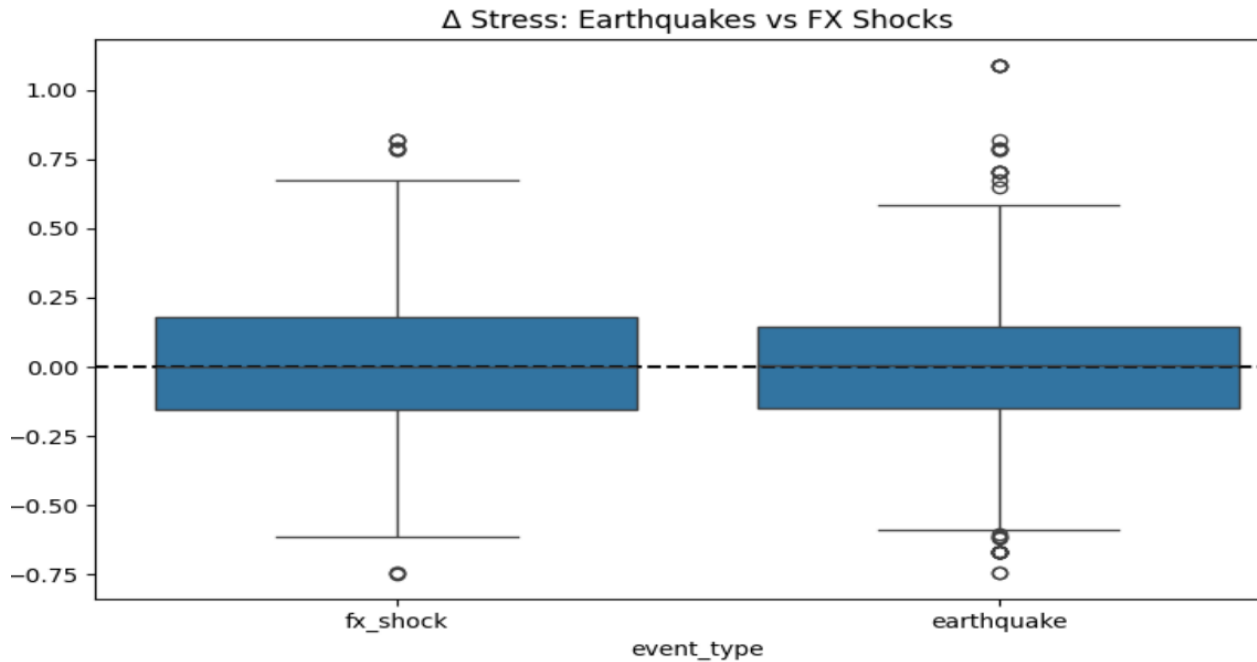


Figure 4. Delta-stress distributions by event type: FX shocks vs earthquakes.

Comparing event types helps interpret whether responses differ by shock mechanism. Both earthquakes and FX shocks are associated with wide stress-response distributions, with similar central tendencies. This supports the view that both event types can elevate stress, but regional heterogeneity is limited when aggregated across events.

5.4 Event Timeline View

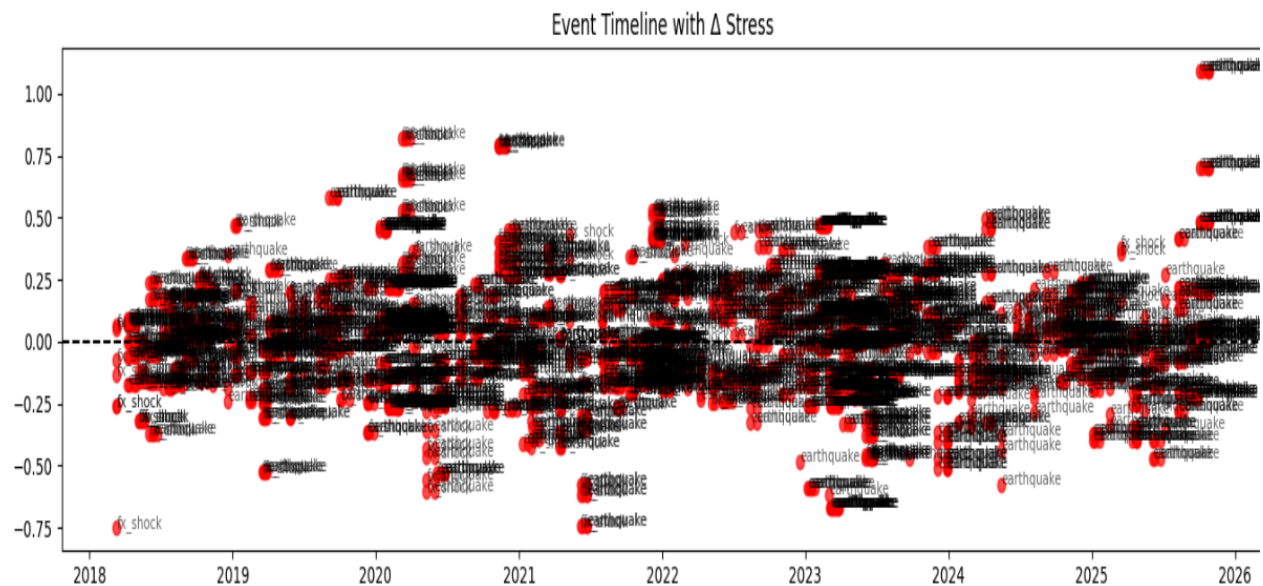


Figure 5. Event timeline with delta-stress values annotated across 2018–2025 (dense plot illustrates event frequency and variability).

The event timeline emphasizes that delta-stress varies widely across individual events, and many events occur close together in time. Overlapping shocks (e.g., clusters of earthquakes or periods of FX volatility) complicate clean attribution and motivate careful interpretation of statistical results.

6. Statistical Testing

To formally test whether regions respond differently on average, the project uses a one-way ANOVA on delta-stress with region as the factor.

Null hypothesis (H_0): All regions have the same mean delta-stress.

Alternative (H_1): At least one region differs.

Reported result: ANOVA p-value = 0.0886.

At the conventional $\alpha = 0.05$ threshold, $p = 0.0886$ implies insufficient evidence to reject H_0 . In practical terms, the data does not support strong, systematic regional differences in post-event stress changes. The finding is consistent with a synchronized national stress response, where shock impacts are broadly shared across regions rather than localized to distinct behavioral patterns.

7. Phase 3 – Machine Learning: Clustering Regional Stress Profiles

Phase 3 extends the analysis with unsupervised learning. Because there is no ground-truth label for “high-stress” versus “low-stress” regions, clustering is appropriate for exploratory pattern discovery.

Method: K-Means clustering ($k = 3$)

Features: mean stress index, maximum stress index, and standard deviation of the stress index (computed per region)

Preprocessing: StandardScaler normalization

Evaluation: Silhouette score (higher indicates better cluster separation)

7.1 Interpretation of Clusters

Clusters can be interpreted as broad profiles such as:

- Higher baseline and higher volatility regions (more reactive overall)
- Moderate baseline with occasional spikes (intermediate profile)
- Lower baseline and lower volatility regions (more stable profile)

Given the ANOVA result, clusters are expected to be weakly separated; nonetheless, clustering provides a complementary view by summarizing multi-dimensional stress behavior instead of focusing only on average delta-stress around events.

7.2 How to Reproduce the Clustering

- 1) Build the regional feature table (one row per region).
- 2) Standardize the feature columns (z-scoring across regions).
- 3) Fit K-Means with $k = 3$ and export region $\rightarrow$ cluster assignments.
- 4) Visualize clusters (e.g., mean vs max stress colored by cluster).

7.3 Visual Results: Regional Stress Response Clusters

Figure 6 illustrates the results of the K-Means clustering applied to engineered regional stress features. Each marker corresponds to one of Türkiye's seven geographical regions, positioned according to mean and maximum stress index values. Colors represent cluster assignments, revealing latent stress-response profiles. Regions grouped within the same cluster exhibit similar overall stress reactivity, while separation across clusters reflects differences in baseline stress and peak responses. This visualization provides an intuitive, multivariate complement to the event-window and ANOVA analyses.

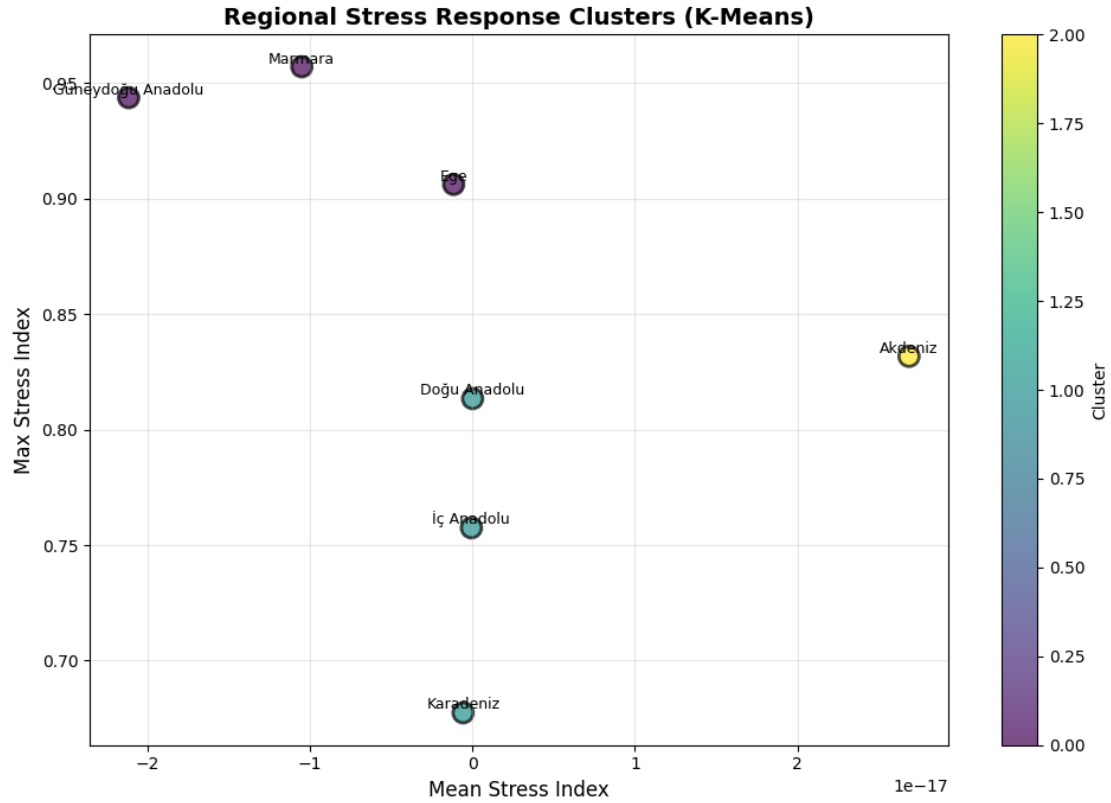


Figure 6. K-Means clustering of regional stress-response profiles based on mean and maximum stress indices. Marker colors indicate cluster membership.

8. Challenges and Limitations

- Google Trends normalization: interest scores are relative (0–100), so comparisons can be affected by query scaling and baseline differences.
- Sparse/noisy provincial data: smaller provinces may exhibit more noise; aggregation helps but can hide local variation.
- Causality: correlations around event windows do not prove causal links; multiple events can overlap in time.
- Event definitions: earthquake magnitude/depth thresholds and FX return thresholds are design choices; sensitivity analysis may change results.
- Window length: a 30-day window may miss longer-term stress impacts or delayed responses.

9. Future Work

- Lead-lag analysis: compute time-to-peak stress per event and region to directly measure reaction speed.
- Event-specific models: separate pipelines for earthquakes vs FX shocks; include interaction terms.
- Additional macro events: incorporate CPI release dates, central bank interest-rate decisions, and election timing.
- Panel regression: region fixed effects and controls (e.g., population weighting, socioeconomics) for more rigorous inference.
- Validation: compare Trends-based stress with external indicators (surveys, mental health service utilization, hospital admissions).

10. Reproducibility and Repository Structure

The repository is organized as a sequence of scripts producing CSV outputs. A typical run order is:

- 1) collect_trends_provinces.py
- 2) build_region_stress_index.py
- 3) build_event_dates.py
- 4) analyze_region_stress.py
- 5) plot_region_stress_analysis.py

Key outputs include:

- region_stress_index_weekly.csv
- event_dates.csv
- event_region_stress_panel.csv
- visualization figures in the outputs/ or figures/ directory (depending on configuration).

AI Assistance Disclosure

ChatGPT was used to assist with report structuring and wording. All implementation choices, data processing, analysis execution, and interpretation of results were performed by the author.